

# Rocket Project

John P. Dang, Cael A. Hand, Sahaj M. Patel  
Georgia Institute of Technology, Atlanta, GA, 30332

## Abstract

This rocket report will describe the specific measurements and developmental process of a single-staged, four-fin, A8-3 motor rocket that was used to achieve an objective height of 125 feet. The rocket also adhered to safety requirements and the integrity of its payload. Teamwork and resources, such as the Aero Maker Space and Open Rocket, that helped design, simulate, and modify the rocket, which played a crucial role in the rocket's performance and launch, will also be discussed. Our result of the project was a model rocket that reached a maximum height of around 94ft, which was an underestimation by our simulated prediction on Open Rocket. We consider this to be the fault of having a stability too high or our center of gravity being too far up our rocket. This caused the rocket to steer into the wind during flight, taking it off its intended path. Furthermore, our parachute did not fully deploy, as a result of not having enough wadding paper to block it from the heat of the engine. Our conclusions for improvements include lowering where we placed most of our weight to lower our center of gravity, as well as adding more wadding paper to ensure the recovery system works exactly as intended.

## Nomenclature

CM = center of mass

CG = center of gravity

CP = center of pressure

## I. Introduction

The main objective of this rocket project was to plan and construct a model rocket that achieved an apogee as close as possible to 125 feet without going over. In addition, we had to design a new component, the payload bay, that would carry the PerfectFlite Pnut altimeter during the rocket launch. The overall method for achieving these objectives consisted of using the Open Rocket software to simulate different variables and parameters for rocket flight as well as using Tinkercad and Solidworks CAD software for 3D modeling and printing potential designs. The general outline throughout this paper depicts every crucial step throughout the rocket project process, including the model rocket kit selection, design modifications and configurations, weight and balance, design modification processes, tests and evaluations, and conclusions.

## II. Model Rocket Kit Selection

We selected the Estes Viking Kit for our Rocket Project. This kit provided us with all the essential components to construct and complete a successful rocket launch. This kit comprised a body tube, nose cone, fins, engine block, streamer, and a shock cord in which each element contributed significantly to the final rocket design. Instead of using the provided streamer, our group purchased the Estes-branded nine-inch parachute to use as our recovery system.

### **III. General Overview of the Design Modification**

The overall dimensions of our final rocket consisted of a length of 39.1 cm and a maximum diameter of 2.3 cm. In addition, the overall mass of the rocket with no motors was 55.5 grams, and the overall mass with the A8-3 motor, a 13.8-gram motor, was 69.3 grams. The overall rocket configuration, from top to bottom, involved a nose cone with a tube coupler, a 3D-printed payload bay with another coupler, a body tube, and four fins.

The main modification in our rocket was the formation and creation of a payload bay to contain the altimeter during flight. This was a critical part of the design modification process because if the altimeter did not fit within the payload bay, there would be no way to record the height of the rocket accurately. The dimensions of the altimeter given to us were 6.35 cm (length) x 1.5 cm (width) x 1.14 cm (height). To provide enough clearance for the payload to enter and exit smoothly, we ensured to add enough centimeter difference to the inner diameter of the payload bay from the altimeter. The final dimensions of the payload bay were a length of 8 cm, an inner diameter of 2.1 cm, and an outer diameter of 2.3 cm.

Another modification in our rocket was the increase in mass of the nose cone. The original mass of the nose cone was 4.55 grams. To increase the mass of the nose cone, we inserted 1X7 mm nuts. These nuts fit perfectly within the nose cone, and we added seven of these nuts. Each nut has a mass of 2.76 grams and seven of these result in a mass of 19.32 grams. The addition of mass to the nose cone was crucial because this allowed the CG of the rocket to be in front of the CP, which resulted in a stable rocket.

The motor we chose was the A8-3. This motor fits perfectly inside the provided body tube from the model rocket kit. Therefore, no modifications were needed for the engine mount inside the body tube. To insert the engine block, we used a super glue bottle with a long, skinny nozzle to apply the glue at the desired location. Then, we immediately fixed the engine block onto the applied super glue. With this in place, the motor could snugly fit into the engine mount.

Furthermore, no modifications were made to the body tube and fins. These components were used directly from the model rocket kit with no alterations. The body tube consisted of a length of 22 cm, an inner diameter of 1.8 cm, and an outer diameter of 2 cm. Inside the body tube, the parachute, shock cord, engine block, and motor were housed. For the fins, the standard kit fins had a sweep length of 4.72 cm, a sweep angle of 72.2 degrees, and a height of 3.26 cm. Four of these fins were used and attached to the bottom of the rocket, each equally spaced apart from each other every 90 degrees.

### **IV. Weight and Balance**

Table 1: Weight and Location of Important Components

| Component Name      | Mass (g) | Weight (N) | Location (cm) |
|---------------------|----------|------------|---------------|
| Nose Cone (w/ mass) | 23.87    | .234       | 0             |
| Payload Bay         | 11.27    | .11        | 5.5           |
| Altimeter           | 7.3      | .072       | 10            |
| Body Tube           | 8.93     | .088       | 15.3          |
| Recovery System     | 2.14     | .021       | 15.7          |
| Launch Lug          | 0.071    | .001       | 30.4          |
| Engine Block        | 0.3      | .003       | 30.3          |
| Fins                | 3.83     | .038       | 35.05         |
| Motor               | 13.8     | .135       | 30.4          |

The measurements shown in Table 1 were collected from the Open Rocket software. Overall, the mass and weight of the model rocket are 69.3 grams and .679 Newtons, respectively. The center of mass of the rocket, the balancing point, was relatively around 18.3 cm from the tip of the nose cone. The center of pressure is located 34.8 cm from the nose cone as well. The difference between the center of gravity and the center of pressure is 16.5 cm. This led to a stability constant of 7.15 calibers, which is a very high stability.

$$CM = \frac{1}{m_{tot}} \cdot \sum m_1 x_1$$

$$CM = (23.87)(0) + (11.27)(5.5) + (7.3)(10) + (8.93)(15.3) + (2.14)(15.7) + (0.0071)(30.4) + (0.3)(30.3) + (3.83)(35.05) + (13.8)(30.4)$$

$$m_{tot} = 69.3 \text{ g}$$

$$CM = \frac{868.27934 \text{ g}}{69.3 \text{ g}} = 12.5$$

We measured the mass of each component using a scale. In the end, the overall mass of the entire rocket accumulated to be 70.2 grams. This transfers to a weight of 0.688 Newtons. Although we measured this, we trusted the Open Rocket measurements because there would be no discrepancies or minor alterations with the scale. This was also true for the center of mass. The calculated center of mass, shown above, was 12.5 cm, but the center of mass within the Open Rocket software was 16.5 cm. Again, we trusted the Open Rocket calculations.

## V. Design Modification Process

The team's first decision was whether to choose the model provided by the kit or create new parts from 3D printing. We decided to choose the base model given by the kit for the simplicity of only having to be concerned with printing the payload bay parts. Then we

recorded measurements of each base component. A disadvantage of only working with the base kit is that we had to work around those measurements, which affected our weight decisions.

The next step was to import all the measurements we took into Open Rocket. At this point, we started thinking about variables and constants. The constants we couldn't change for our Open Rocket simulations were the components of the base kits: the main body tube, engine block, nose cone, altimeter, shock cord, and launch lug. The variables that we could change to affect our vertical height and velocity were the measurements/materials of the payload bay, the amount of mass added, the motor used, and the number of fins.

First, we decided on the material and the size of the payload bay since this was one of the biggest factors. The initial choices out of the Open Rocket presets were PVC ( $1.39 \text{ g/cm}^3$ ), carbon fiber ( $1.78 \text{ g/cm}^3$ ), PETG ( $1.27 \text{ g/cm}^3$ ), or polystyrene PS ( $0.905 \text{ g/cm}^3$ ). Our final decision was to make it out of polystyrene PETG ( $1.27 \text{ g/cm}^3$ ) due to its lightweight properties and manufacturing ease. Our next critical design decision was to choose our motor. After inserting into Open Rocket our payload bay and its necessary diameters to fit into the body tube, we simulated the rocket with three fins and with all four motors to compare the maximum vertical distance with each other. We found that 1/2A3T and 1/2A6-2 reached altitudes below the target height. The stronger motors, A8-3 and B4-4, reached heights greater than the target apogee. The simulations were also done with a four-fin configuration and each motor's maximum height slightly decreased from its three-fin configuration.

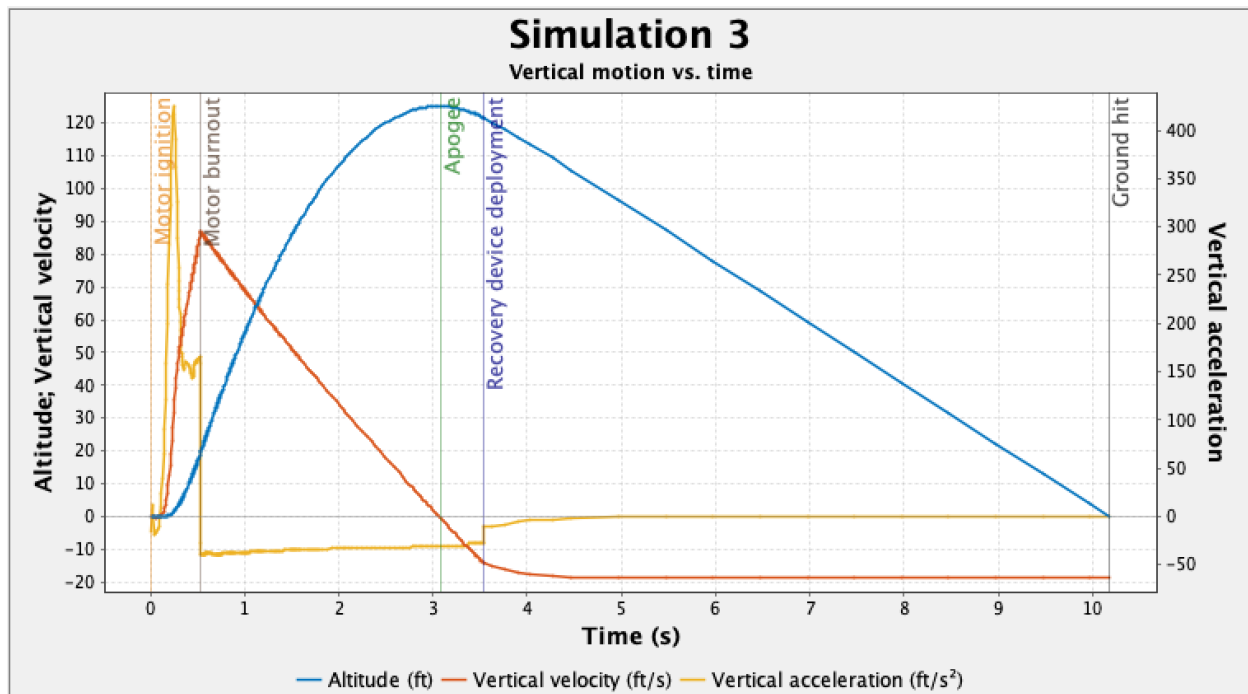
The four-fin configuration gave us higher stability and it was easier to put on the main body since there was a ninety-degree angle between each fin, so we chose that option. Once the rocket's base components were all decided, the next developmental step was to modify the rocket's components to reach the target height. The variables we could change were too limited for the motors that fell short of the target height. Any changes we could make would have improved the difference slightly. So, we decided to use the more powerful motors and add mass to decrease their height differences from the target height. Between the two options, we chose A8-3 because its simulated maximum height of 240.85 feet had a smaller difference from the target altitude compared to the B4-4 motor.

Initially, we placed batteries inside the nose cone to reduce the maximum height. However, Professor Zaidi recommended using objects like washers and nuts instead due to safety concerns and the rocket's stability since the battery was loose in the nose cone. While taking recommendations from Professor Zaidi, we 3D printed three payload bays and a tube coupler. The first payload diameter was too small which made altimeter accessibility difficult, and the coupler wasn't a tight fit either. The second design was better at holding the altimeter but was too long according to Professor Zaidi and the coupler was a loose fit. After shortening the length of the payload bay, we used tape to increase the tightness between both ends of the payload bay.

Afterwards, we realized the landing velocity of  $\sim 85 \text{ ft/s}$  was too high and there was a high possibility of damaging the payload. To counteract the high landing velocity, a 9-inch Estes 2268 printed Parachute was bought from Amazon to replace our streamer. At this point, the design and

manufacture of the rocket was complete. The only last-minute decision to modify the design was to tape the nose cone and the payload bay for a more secure hold. During the launch day, we used only two wadding papers, which was enough to prevent major damage to our recovery system. Referenced below is Figure 1, the culmination of all our design modifications to the rocket and its effect on the simulated rocket path. The maximum apogee predicted was around 124.43 ft, but this didn't include small glue components of mass that we knew we would add to the rocket's assembly.

Figure 1: Final Simulated Path of Rocket



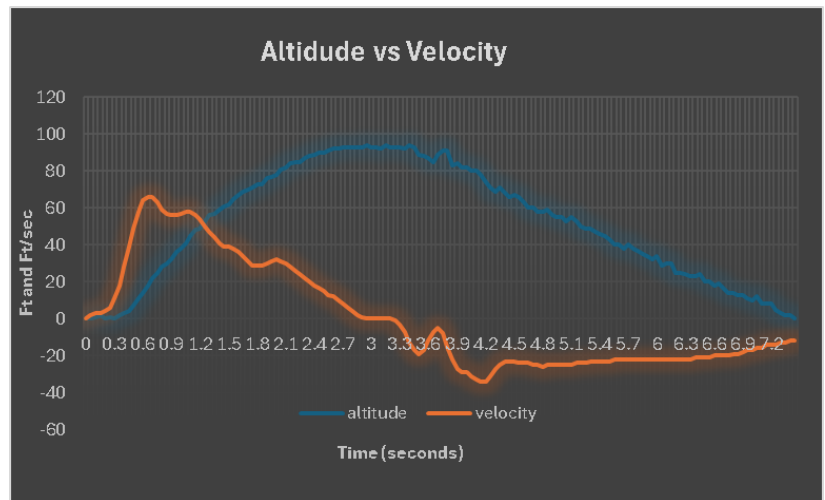
## VI. Test and Evaluation

The day of the rocket launch was Thursday, October 3, around 10:20 am. The weather was very sunny with low winds and few clouds. To prepare for our launch we watched the instructional video posted that explained how to fold our parachute into our rocket, insert the wadding paper (we used 2 squares), and prepare the engine by attaching the ignition wires. We also had to wrap 2 pieces of tape around our engine to make it slightly more snug in the body tube. The mass of all these final components put our measured rocket weight a few grams over our predicted  $m_{tot}$  to around 70.2g. We did this all on the morning of launch and then waited in line to get approved for launch. The TA told us that our CG was a little higher than most teams, and to tape our nosecone to the payload for insurance it stays on, and then we were given the good to go. Upon placing our rocket on the launch pad, we noticed that since the launch lug was attached to the 18mm diameter, the launch rod bent the rocket a little since the payload bay was 3mm wider. We then attached the ignition wires to the starters and were ready for launch. After a countdown of 5, we had liftoff.

Figure 2: Path of Rocket



Figure 3: Plot of Altitude and Velocity



Our rocket was launched, and we observed it heading straight upward for about 2 seconds. At that point, the rocket started to lean to one side and lost a lot of upward momentum. The initiation of the recovery system deployed at around 3.5 seconds when the rocket was practically horizontal and can be seen in Figure 2 where the spike in velocity is around that time. When the recovery system released the parachute, we observed that it didn't fully open, though it did slow the rocket down to around -12 ft/s when it hit the ground, which was around the suggested impact speed. The rocket also landed within 10 feet of the launch pad.

After observing our rocket's path through the air, we knew it probably did not reach the full 125ft as it started tilting sideways in the air. Our guess for our rocket's maximum height was around 115 feet, but Professor Zaidi read our altimeter data and told us our apogee was around 94 feet. This was consistent with our observation of undershooting the target height.

Figure 1 represents the predicted flight in Open Rocket, and Figure 3 represents the actual flight from the altimeter. The actual flight compares to the Figure 1 model in terms of descent. The descent matches with what happened in reality in which the parachute significantly decreased the landing speed to be below 20 feet per second. On the flip side, the actual flight did not compare to the Figure 1 model in terms of vertical height achieved. The predicted height was 125 feet whereas the real height was 94 feet. However, since our rocket began to tilt and move horizontally, this horizontal movement could have contributed to the vertical height, increasing the real height. These were the major similarities and differences of the actual and predicted rocket flight as well as the comparison to the model.

Figure 4: Rocket Post-Flight



For the evaluation of our rocket, we first noticed that the parachute did not fully deploy because the engine had burnt and melted a portion of it together, which didn't allow it to fully open during descent. We think this was due to only using two squares of wadding paper between our engine and recovery system. Next in our evaluation was the tilting of the rocket during its flight. We overheard Professor Zaidi telling another group that had a stability of over 8 in Open Rocket, that too much stability would result in the vehicle trying to point to the wind. Our rocket having a stability of around 7 fell victim to this as well. If we wanted a smaller stability, we would have needed to lower our CG closer to our CP. This might involve lowering where we placed our weights or redesigning

with a lower-power engine that doesn't require a lot of added mass so that it would be more aerodynamic. In evaluating our modifications, the altimeter was in perfect condition upon recovery. The data also shows that the temperature of the altimeter never increased significantly, so it was protected well by our payload bay. We hypothesize the total distance traveled to the apex of the flight of our rocket was close to 125 feet, with 94ft vertical distance, and 30ft horizontal distance. If this is true, then our Open Rocket prediction was accurate. All pieces that were meant to stay attached did, and the retrieval of the altimeter was an easy process.

## **VII. Conclusion and Recommendations for Further Design Improvements**

Our final design was a four-fin, single-staged, and ogive-shaped nose-coned rocket with a total length of 39.1 cm and a maximum diameter of 2.3 cm. The rocket used an A8-3 motor with a 9-inch parachute. The center of pressure is located 34.8 cm from the nose cone of the rocket. The center of gravity is located 18.3 cm from the rocket. The Open Rocket simulated a maximum altitude of 124.43 feet with a landing velocity of -18.55 feet.

The Open Rocket simulation overestimated our height. The simulation estimated an apogee of 124.43 feet, but the recorded apogee of the actual flight was 94 feet. However, the Open Rocket simulation correctly predicted the descent rate of rocket. The rocket descended at a safe speed which allowed the altimeter to be safely protected, which our data on Open Rocket predicted.

Our rocket project is considered a success. We completed our objective to design, plan, and construct a model rocket that kept the altimeter safe inside a payload bay. Along the way, we made many modifications to achieve this goal, went back on many designs, and eventually ended up with a product that was given the go-ahead for launch. After reaching an apogee of 94ft, we realize there is room for improvement. Further design improvements would include lowering

the position difference between the CP and CG, resulting in lower stability so the rocket wouldn't point toward the winds. Also, we would add 3 or 4 pieces of wadding paper instead of 2, in hopes that the parachute would deploy properly, better protecting the altimeter.

### **VIII. Teamwork and Collaboration**

There are three members within this group: John P. Dang, Cael A. Hand, and Sahaj M. Patel. Each member played a significant role in the success of the rocket project. Cael was mainly responsible for the safekeeping of the rocket, measurement collection, and assembly of the rocket. Sahaj handled most of the Open Rocket simulations and 3D printing of components at the AE Maker Space. John was also responsible for assembly, handling data collection from the simulations, and research for the initial design. Meetings were mainly held during and outside of work periods, and communication was consistent throughout the duration of the project with the use of a group chat. Most of the measurements and Open Rocket files were shared through either the group chat or Outlook emails. The rocket report is also a collaboration between all three members and discussions for major design decisions were always held in person between the members for effective communication.

### **IX. Background Research**

Our group researched numerous amounts of content before we planned and constructed our rocket. The first, and most important, step of the research process was to find the dimensions of the altimeter. We went to the official PerfectFlite website [1] and determined the dimensions. This was very critical because this would allow us to create the payload container with accurate dimensions. If the altimeter did not fit inside the payload bay, no height would be recorded. Next, we researched the stability constant of a rocket. The CM must be in front of the CG for a rocket to be classified as stable.

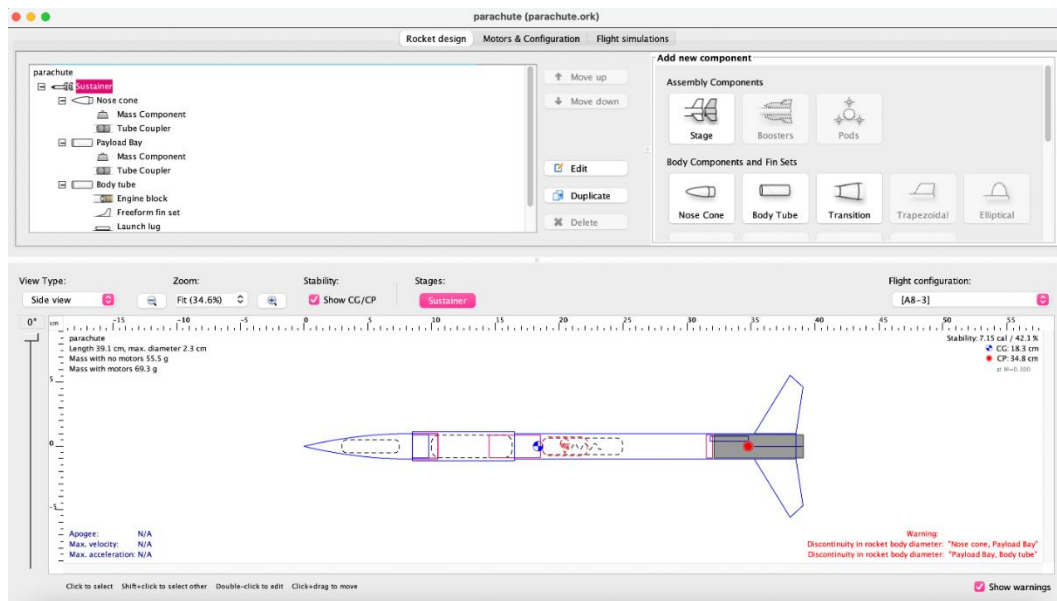
In addition, we found that a standard range of stability should be that the CM must be a body tube-length in front of the CG [2]. In addition, it was determined that the stability should be in a range of 5-6 calibers. This information allows for the rocket to be stable. In our rocket design, we achieved the first part of this, having the CM a body tube-length in front of the CG as shown in the Appendix. However, our stability was 7.15 calibers, which is higher than the standard range. This high stability led to our rocket being top-side heavy and leaning towards the wind during flight.

Another research element was to figure out how to increase or decrease the height based on a change in different variables. Two efficient methods we found to decrease the vertical height of a flight were to either add mass or increase the length of the payload bay. Two efficient methods we found to increase the vertical height of a flight were to either remove mass or to decrease the length of the payload bay. We experimented with these parameters within the Open Rocket simulation to see which configuration would result in a vertical height closest to an apogee of 125 feet. The picture in the Appendix reflects this final configuration.

Another aspect of the research was to determine the weather and wind during the time and day of the launch. Using a weather source, AccuWeather, we determined that there would be slight north-to-south winds during the time and day of the launch [3]. This information was needed to determine if we needed to overestimate or underestimate the height of the Open Rocket software.

Finally, we researched the number of pressure holes and their sizes to make sure the altimeter could properly read the pressure change. The Pnut User Manual from PerfectFlite gave us the diameters depending on the payload length, their relative positioning to the nosecone shoulder, and the number of holes we should have to avoid irregularities [4]. The planned final design had two holes with  $\sim 0.012$  mm in diameter for each hole. Overall, all the research performed was critical in the making of a successful rocket launch, and all the research had a profound impact on the design thought and planning process.

## X. Appendix



## XI. References

[1] PerfectFlite, "Perfectflite Home Page," *PerfectFlite Home Page*, Available:

<http://www.perfectflite.com/>.

[2] Firnau, A., "Model Rocket Stability Prediction," *SpaceCAD model rocket design software*,

Available: <https://www.spacecad.com/posts/model-rocket-stability-prediction/>

[3] AccuWeather, "AccuWeather Georgia Tech GA", Available:

<https://www.accuweather.com/en/us/georgia-tech/30318/october-weather/2625834>

[4] PerfectFlite, "Pnut User Manual," Available:

<http://www.perfectflite.com/Downloads/Pnut%20manual.pdf>